

Product Requirements Document

Ariga Justine Download Manager (AJDM)

Document	AJDM Product Requirements Document (PRD)
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Author / Owner	Ethan (founder, lead engineer)
Companion document	architecture.md (technical design; requirement IDs here map to it)
Platforms	Windows 10/11, macOS 12+, Linux (Ubuntu 22.04+, Pop!_OS, Fedora)
Primary technology	Java 25, JavaFX, yt-dlp (extraction only), ffmpeg (mux only)

“AJDM carries two names: Ariga and Justine. It is built in memory of my brother, and it is built to last.”

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1. Executive summary

AJDM is a cross-platform desktop download manager that does two things exceptionally well. First, it downloads any HTTP(S) file as fast as the user's connection and the server allow, using adaptive parallel connections with live rebalancing. Second, when given a video or playlist link (YouTube and every site supported by yt-dlp), it lists every available format and quality, including audio-only and video up to 4K (2160p), and lets the user choose exactly what to download.

The market leader, Internet Download Manager (IDM), is Windows-only, closed-source and built on a two-decade-old engine. AJDM's objective is to **outshine IDM**: run on Windows, macOS and Linux, expose its internals (segment map, per-connection speeds), and beat IDM's measured download time in a published, repeatable benchmark.

Brand and purpose. AJDM stands for *Ariga Justine Download Manager*. The name and logo honour the author's late brother and the author himself. The photograph in the logo is part of the product's identity and appears in the About screen, the installer and the website. *My brother. My legacy.*

1.1 Product vision

A fast, honest, beautiful download manager that works on every desktop, built with care and meant to outlive its creator's first version.

1.2 One-page summary

Item	Decision
Target users	Students and professionals on 4G/5G and broadband who download large files and videos; creators who need 4K sources; Linux users who have no IDM.
Differentiators	Cross-platform; adaptive speed engine with rebalancing; full format picker (video/audio/4K); open diagnostics; resume that survives crashes; data-saver and off-peak scheduling.
Engine approach	Own Java engine (virtual threads, HTTP range requests, positional writes). yt-dlp used only to discover formats and URLs. ffmpeg only to mux video+audio without re-encoding.
Success definition	Median time-to-complete at or below 0.95x IDM on at least 70% of the benchmark scenarios and never worse than 1.10x in any; zero corrupted files in chaos testing.
v1.0 timeline	About 20 weeks of part-time solo work (see section 14).

2. Problem and opportunity

2.1 Problems today

- **No good cross-platform IDM equivalent.** IDM is Windows-only. Linux and macOS users rely on slower or less polished tools.
- **Browsers download over one connection** and handle interruptions poorly. On mobile-network links and CDNs that throttle per connection, a single stream leaves bandwidth unused.
- **Video quality selection is clumsy.** 4K video on most platforms is shipped as separate video and audio streams that must be downloaded and merged. Users want a clear picker and a one-click result.
- **Fragile resume.** Many tools cannot resume after a crash, an IP change or an expired link, wasting data. In regions where data is bought in bundles, wasted bytes cost real money.
- **Closed behaviour.** Users cannot see why a download is slow. IDM does not show per-connection performance.

2.2 Opportunity

IDM's core ideas (segmented downloads, dynamic segmentation) are well understood. What IDM has is two decades of tuned thresholds. A modern engine with continuous measurement, per-host learning, connection recycling and a purpose-built test and benchmark harness can match that tuning faster than it was originally achieved, and add cross-platform reach.

2.3 Honest physical limit

No downloader can exceed the user's link or the server's capacity. For example, a 20 GB 4K video on a 10 Mbit/s link needs about **4.4 hours** at perfect line rate. AJDM's promise is to **saturate the available link whenever the server permits**, to recover from problems without re-downloading, and to beat IDM where engines differ. Marketing claims will always be benchmark-backed and state the scenario.

3. Goals, non-goals and success metrics

3.1 Goals

ID	Goal	Measure
G1	Be faster than IDM on realistic networks	Benchmark matrix (section 12): median time <= 0.95x IDM in >= 70% of scenarios; <= 1.10x in all
G2	Show every format and quality of a media link	100% of non-DRM https formats reported by yt-dlp are listed with codec, size, fps, resolution
G3	Never corrupt or lose data	0 hash mismatches in 1000 chaos runs; 200/200 kill-9 resumes bit-exact
G4	Work everywhere	Same feature set on Windows, macOS, Linux; CI green on all three
G5	Feel trustworthy and clear	Time from paste-URL to first byte <= 1.5 s for direct links; clear error messages for all catalogued error codes
G6	Honour the name	About screen, installer, website carry the dedication and logo

3.2 Non-goals for v1.0

- FTP/SFTP, BitTorrent and magnet links.
- DRM circumvention of any kind. DRM-protected content is detected and refused.
- Mobile apps (Android/iOS).
- Cloud sync or accounts. AJDM is local-first with no server component.
- Re-encoding or transcoding media (only lossless muxing).
- A built-in browser or media player.

3.3 Key performance indicators

KPI	Target	How measured
Link utilisation when server is unthrottled	>= 90% of measured link capacity	ajdm-bench, LAN and CDN scenarios
Link utilisation when server caps each connection at C	>= 95% of min(link, N x C)	ajdm-bench, nginx limit_rate scenario
Resume success after interruption	>= 99%	chaos + kill-9 suites
Crash-free sessions	>= 99.5%	opt-in crash reports, beta cohort
Format list time for one YouTube video	<= 6 s typical	stopwatch in bench (depends on yt-dlp)
Cold start (UI visible)	<= 2 s on 8 GB RAM laptop	scripted launch timing with AppCDS
Idle memory (RSS)	<= 200 MB	OS process stats

4. Users and scenarios

4.1 Personas

Persona	Profile and need	What AJDM must do for them
Amani, the student	University student in Kenya on a 4G data bundle; downloads lecture videos, course ISOs and software; cannot afford wasted data; often has night bundles.	Resume without re-downloading; data-saver mode; schedule downloads in off-peak windows; clear remaining-size estimates.
Wanjiru, the creator	Edits video; needs 4K source clips and clean audio; downloads whole playlists.	Format picker with presets (Best up to 4K); playlist select-all; lossless mux; predictable file naming.
Kevin, the Linux developer	Runs Linux; downloads large images and datasets; wants automation.	Native Linux package; CLI (`ajdm add`); headless daemon; speed and segment diagnostics.
Grace, the occasional user	Just wants big files to download reliably and faster than the browser.	Browser extension that takes over downloads; zero-configuration defaults.

4.2 Primary user stories

ID	As a...	I want to...	So that...
US-1	user	paste a link and see the download start within seconds	I do not wait or configure anything
US-2	user	paste a YouTube link and choose from every quality up to 4K, plus audio-only	I get exactly the file I need
US-3	user	paste a playlist and select which videos and what quality for all	I can batch many videos at once
US-4	user	have downloads continue after a crash, restart or network change	I never start over
US-5	user	see why a download is slow (connections, per-connection speed)	I can understand and trust the tool
US-6	user	limit speed and schedule downloads for night hours	I can protect my bundle and my daily internet use
US-7	user	click Download with AJDM in my browser	I do not copy links by hand
US-8	user	open About and see the story behind the name	I know what AJDM stands for

5. Brand and identity

The logo combines the AJDM initials with the photograph of the two brothers and a download badge. It is the product's mark and is used wherever AJDM appears. The photograph is never altered, recoloured, or cropped to remove either person.

5.1 Logo assets

Asset	Use
Horizontal lockup (assets/ajdm_logo.png)	Documents, website header, About screen, installer banner, README.
Square icon (assets/ajdm_icon.png)	App icon, installer icon, extension icon (export 16/32/48/128), tray icon (simplified).

5.2 Colour tokens

Token	Hex	Usage
Legacy Navy	#0B1F3A	Backgrounds, primary text on light, headers
Memory Gold	#E8B04A	Accents, tagline, focus rings, highlights
Flow Teal	#1FB5A8	Progress, active states, the letter D in the wordmark
Mist	#EEF3F8	Panels and zebra rows on light theme
Slate	#5B6B82	Secondary text

5.3 Rules

- Minimum clear space around the logo equals the height of the letter A.
- Tagline wording is fixed: *My brother. My legacy.*
- Wordmark letters: A, J, M in white (or navy on light); D in Flow Teal.
- On very small sizes (tray, 16 px favicon) use a simplified glyph: the teal download badge on navy.

5.4 Dedication copy (About screen)

AJDM stands for Ariga Justine Download Manager. It carries both our names. I built it in memory of my brother, so that something of us keeps moving forward, one byte at a time.

6. Scope and release plan

Release	Contents
v0.1 Alpha	Probe, single-stream and segmented download, resume, SQLite state, CLI, test server. Internal benchmarks begin.
v0.5 Beta 1	Adaptive controller, rebalancing, per-host learning, benchmark vs IDM, JavaFX main window, queue.
v0.8 Beta 2	Media pipeline (formats, presets, playlists, mux), scheduler, settings, tray.
v1.0	Browser extension, installers for all three OSes, signed builds, About/dedication, documentation, public benchmark report.
v1.1	HLS/DASH fragment downloads, subtitles/thumbnail/metadata embedding, auto-update, Swahili UI, mirrors.
v2	Metalink, FTP/SFTP, HTTP/3 experiments, remote control, plugin API.

7. Functional requirements

Priority uses MoSCoW: **M** Must (v1.0 blocker), **S** Should, **C** Could. Release column shows the earliest target. Each requirement has testable acceptance criteria (AC).

7.1 Transfer engine (ENG)

ID	Pri	Rel	Requirement and acceptance criteria
ENG-01	M	0.1	**Probe.** Determine size, range support, validators (ETag/Last-Modified), final URL and filename using HEAD, falling back to GET Range 0-0. AC: correct results on a corpus of >= 100 URLs across >= 20 hosts; probe completes in <= 1.5 s on a healthy server.
ENG-02	M	0.1	**Segmented parallel download** via HTTP Range. Start with 4 connections; user max default 16, hard cap 32. AC: output SHA-256 equals source on all test servers.
ENG-03	M	0.5	**Adaptive concurrency.** Slow-start doubling then hill-climb, link-bound detection, multiplicative decrease on 429/503/resets. AC: on a server capping each connection at C, throughput reaches >= 95% of min(link, N x C) within 10 s.
ENG-04	M	0.5	**Work stealing and laggard handling.** Idle workers steal from the segment with the largest ETA; laggards (< 0.35x median for 3 s) lose their upper half; < 0.10x median for 5 s recycles the connection. AC: with one deliberately slowed connection, total time is within 10% of the same test without the slow connection.
ENG-05	M	0.1	**Direct positional writes** into a preallocated .ajdmpart file; no merge step; atomic rename at the end. AC: peak extra disk usage <= file size + 64 MiB.
ENG-06	M	0.1	**Pause/resume and crash recovery.** Checkpoint every 5 s after fsync. AC: kill -9 at random points x 200 runs, all complete bit-exact; at most 5 s of data re-downloaded.
ENG-07	M	0.1	**Source-change detection** with If-Range and validators. AC: when the server changes the ETag mid-download, AJDM pauses and asks, never splicing two versions.
ENG-08	M	0.1	**Fallback to single stream** when ranges are unsupported or length unknown; UI states that resume is unavailable. AC: tested against a server returning 200 to ranged requests.
ENG-09	M	0.1	**Retry policy** with exponential backoff, full jitter, per-window max 10, classification by status code. AC: all conditions in the architecture retry table covered by automated tests.
ENG-10	M	0.1	**Stall detection.** No bytes for 15 s triggers reconnect. AC: tested with a server that stops sending.
ENG-11	S	0.5	**Speed limiter** (global and per download). AC: measured rate within +/-5% of limit.
ENG-12	M	0.5	**Connection caps** per host (16) and global (64). AC: never exceeded under stress test.
ENG-13	M	0.8	**Proxy support:** system, manual HTTP and SOCKS5, with authentication. AC: downloads succeed through a test proxy.
ENG-14	S	1.0	**Authentication:** HTTP Basic, custom headers, cookies. Digest/NTLM later. AC: protected test server.
ENG-15	S	1.1	**Mirrors:** multiple URLs for one file with load balancing and failover.
ENG-16	S	1.0	**Checksum verification** when a hash is supplied (sha256). AC: mismatch produces AJDM-3005.
ENG-17	M	0.1	**Safe filenames and conflict policy** (rename/overwrite/ask). AC: traversal and reserved-name tests pass on all OSes.
ENG-18	M	0.1	**Disk space pre-check.** AC: refuses to start with AJDM-2003 when free space < size + 64 MiB.
ENG-19	S	0.5	**Per-host learning:** persist best connection count and capabilities; reuse next time. AC: second download from the same host skips slow start.
ENG-20	C	1.0	**End-game duplication** for the last 2 MiB (off in data-saver).

7.2 Media extraction and download (MED)

ID	Pri	Rel	Requirement and acceptance criteria
MED-01	M	0.8	**Resolve media URL** (video or playlist) via yt-dlp JSON output. AC: valid result for a test set of >= 30 videos across >= 5 sites.
MED-02	M	0.8	**Full format table** showing container, resolution, fps, video codec, audio codec, bitrate, size (exact or approximate with a tilde), HDR flag. AC: every non-DRM https format from yt-dlp appears.
MED-03	M	0.8	**Resolution ceiling** default 2160p, with a setting to show up to 4320p and a Show all toggle.
MED-04	M	0.8	**Presets:** Best up to 4K, 1080p, 720p, Audio only (best), Custom. Deterministic selection per the rules in architecture.md section 9.2.
MED-05	M	0.8	**Video-only + audio-only download and lossless mux** with ffmpeg (-c copy). AC: output plays; stream hashes equal inputs; mux of 20 GB completes in under 2 minutes on SSD.
MED-06	M	0.8	**Playlists:** list, select all, range selection, apply one preset or choose per video; unavailable entries skipped with reason. AC: 100-item playlist handled without rate-limit failures.
MED-07	M	0.8	**Expired URL refresh** by re-running extraction for the chosen format; workers continue with the new URL. AC: download that crosses a forced URL expiry completes bit-exact.
MED-08	S	1.1	**HLS and DASH** fragment downloads in parallel.
MED-09	S	1.1	**Subtitles, thumbnail, metadata** download and embedding.
MED-10	M	0.8	**yt-dlp lifecycle:** bundled binary, weekly update check, update on extraction failure, SHA-256 verification, rollback. AC: simulated broken extractor recovers after update.
MED-11	M	0.8	**DRM detection:** show AJDM-4004 and do not offer download.
MED-12	S	1.0	**Browser cookies opt-in** for content the user is entitled to access; never logged or persisted by default.
MED-13	S	0.8	**Display size estimates** for the chosen combination (video + audio) before download starts.

7.3 Queue and scheduler (QUE)

ID	Pri	Rel	Requirement and acceptance criteria
QUE-01	M	0.5	Multiple queues, default *Main*; max concurrent downloads per queue (default 3). Drag-and-drop reorder and priority.
QUE-02	M	0.8	Schedule windows (start/stop time, days); optional strict mode pauses running items when the window closes.
QUE-03	M	0.8	Post-queue actions: none, sleep, hibernate, shutdown, run command, preceded by a 60-second cancelable countdown.
QUE-04	S	0.8	Duplicate detection by normalised URL or ETag + size, with a prompt.
QUE-05	S	0.8	Automatic retry of failed items with configurable attempts and delay.
QUE-06	S	0.8	Categories with default folders (Video, Audio, Documents, Software, Compressed).

7.4 User interface (UI)

ID	Pri	Rel	Requirement and acceptance criteria
UI-01	M	0.5	Main window: list with name, size, progress, speed, ETA, state, queue; filters (All, Active, Completed, Failed); search.
UI-02	M	0.5	Add-URL dialog with clipboard detection, destination folder, queue, start now/later, optional checksum field.
UI-03	M	0.8	Media picker with format table, presets, size estimate and Download button; playlist view with checkboxes.
UI-04	M	0.5	Details pane: segment map (completed vs active ranges), connection count, per-connection speed sparkline, event log.
UI-05	M	0.8	Settings dialog for every setting in architecture.md section 14, grouped and searchable.
UI-06	M	0.8	System tray with quick pause-all, speed limit toggle, open folder; background mode when window closed.
UI-07	M	0.5	Light, dark and system themes using brand tokens.
UI-08	M	0.8	Desktop notifications on completion and on failure with code and action (retry/open log).
UI-09	M	0.5	Keyboard shortcuts for add, pause, resume, delete; full keyboard navigation; accessible names; contrast AA.
UI-10	S	0.8	Drag-and-drop URLs and text files of URLs (bulk add).
UI-11	S	1.1	Languages: English in v1.0, Swahili in v1.1 via resource bundles.
UI-12	M	1.0	About screen with logo, version, licenses, and the dedication text from section 5.4.

7.5 Browser integration (BRW) and CLI

ID	Pri	Rel	Requirement and acceptance criteria
BRW-01	M	1.0	Extension for Chrome, Edge, Brave (Manifest V3) and Firefox that intercepts downloads above a size threshold (default 1 MiB), sends URL, referrer, cookies to AJDM via Native Messaging.
BRW-02	M	1.0	Context menu Download with AJDM for links and media; toolbar button to send current page video to the media picker.
BRW-03	M	1.0	Allow/deny lists and a per-site disable. If AJDM is not running, the browser download proceeds normally (fail-open).
BRW-04	S	1.1	Detect playable media on a page and badge the toolbar icon.
CLI-01	S	0.1	Commands: `ajdm add <url>`, `formats <url>`, `list`, `pause|resume|cancel <id>`, `daemon` (headless). Exit codes mapped to error codes.

7.6 Diagnostics and updates (DIA)

ID	Pri	Rel	Requirement and acceptance criteria
DIA-01	M	0.5	Rolling logs without secrets; per-download trace id.
DIA-02	M	0.5	Per-download speed and connection timeline recorded and exportable as CSV.
DIA-03	S	1.0	One-click diagnostics bundle (redacted settings, logs, host profile, controller decisions).
DIA-04	S	1.1	Signed auto-update for AJDM itself with rollback.
DIA-05	C	1.0	Opt-in crash reporting (default off).

8. Non-functional requirements

Area	Requirement	Target
Performance	Link utilisation; start latency; efficiency	Sections 3.3 and 12; first byte <= 1.5 s for direct links; <= 10% total CPU on 4 cores at 100 MB/s
Reliability	No corruption; recover from crash/power loss	0 corruption in 1000 chaos runs; resume bit-exact; checkpoint every 5 s after fsync
Scalability	Many items	10,000 history rows; 1,000 visible rows at 60 fps; 100 queued downloads
Resource use	Memory and threads	Idle RSS <= 200 MB; 32-connection download RSS <= 400 MB; no thread/handle leaks after 24 h soak
Portability	Same behaviour on all OSes	Windows 10/11 x64 (ARM64 later), macOS 12+ Intel and Apple Silicon, Ubuntu 22.04+/Pop!_OS/Fedora
Security	Local app, hostile web	No TCP listener by default; Unix socket with user-only permissions; TLS always verified; checksum-verified tool updates; no shell strings
Privacy	No tracking	No telemetry by default; opt-in crash reports; cookies and tokens never logged
Accessibility	Inclusive UI	Keyboard-only operation; screen-reader names; WCAG AA contrast; scalable fonts
Usability	Zero-config	A first-time user completes a download with no settings change; every error shows a code and a next step
Maintainability	Solo-friendly	Module boundaries enforced by ArchUnit; >= 80% line coverage on engine and store; CI < 15 min
Compliance	Legal posture	Section 11; no DRM bypass; license notices for bundled tools

9. User experience

9.1 Core flows

Flow A: direct file

- Paste or drop a link (or click Download with AJDM in the browser).
- Add dialog fills in filename, size, destination from the probe in about 1 s. User confirms (Enter).
- Details show connections ramping 4, 8, 16 as throughput improves; segment map fills; completion notification offers Open and Show in folder.

Flow B: YouTube video up to 4K

- Paste the video link; AJDM shows Reading formats, then a table grouped as Video + audio, Video only, Audio only.
- Default selection is the preset *Best up to 4K*, with a live size estimate (for example, video ~18.4 GB + audio ~0.3 GB).
- User clicks Download. Two streams download in parallel through the engine, then AJDM merges them losslessly and shows the final file.

Flow C: playlist

- Paste playlist link; list of entries appears instantly (flat listing).
- Select all or a range; choose one preset for all or per video; confirm.
- Entries extract gradually (3 at a time) and queue automatically; unavailable entries are marked Skipped with a reason.

Flow D: interruption

- Network drops: downloads show Waiting for network (not Failed).
- Network returns: auto-resume; expired media links refresh silently; no progress lost beyond the last 5-second checkpoint.

9.2 Screen inventory

Screen	Purpose	Key elements
Main window	Manage downloads	Toolbar (Add, Pause, Resume, Delete, Settings), left filter pane, list, status bar with global speed and active connections
Add URL	Single/bulk add	URL field, detected name/size, folder, queue, start now/later, checksum, Advanced (headers, cookies, connections)
Media picker	Choose formats	Thumbnail, title, duration, preset chips, format table, size estimate, container selector
Playlist picker	Batch media	Checkbox list, range input, preset for all, per-row override
Details	Understand speed	Segment map, connection table, speed graph, controller decisions, log
Scheduler	Windows and actions	Queue list, time windows, post-queue action, strict mode
Settings	Configuration	General, Network, Downloads, Media, Browser, Appearance, Advanced
About	Identity	Logo, version, dedication, licenses, links

9.3 Error messaging principles

- Always show a code (for example AJDM-3002) and one plain-language sentence and one action.
- Never blame the user. Prefer Retry, Open log, Copy diagnostics.
- Offline is a state, not a failure.

10. System overview

AJDM runs as one long-lived process hosting the core (engine, scheduler, store). The JavaFX UI, the CLI and the browser native-messaging shim are clients that talk to the core over a Unix domain socket. The core calls origin servers over HTTP/1.1 (HTTP/2 only when learned safe), spawns `yt-dlp` for extraction and `ffmpeg` for muxing, and persists state in SQLite. Full design: **architecture.md**.

Layer	Responsibility	Module
Model	Records, enums, events, error codes	ajdm-model
Network	Transport interface, probe, proxy, cookies	ajdm-net
Engine	Segmentation, workers, controller, writer, limiter	ajdm-engine
Store	SQLite repositories, checkpoints, migrations	ajdm-store
Media	yt-dlp adapter, format model, ffmpeg mux, URL refresh	ajdm-media
Scheduler	Queues, priorities, schedules, post-actions	ajdm-scheduler
Core	Composition root and public service API	ajdm-core
IPC	Socket server and client, native-messaging shim	ajdm-ipc
UI / CLI	JavaFX app; picocli commands	ajdm-ui, ajdm-cli
Bench	Fault-injection server, benchmark harness	ajdm-bench

10.1 Speed strategy (why AJDM can beat IDM)

#	Technique	Expected benefit	Risk
1	Multiple HTTP/1.1 connections per download, tuned live	Defeats per-connection throttling and TCP slow-start limits	Server bans if too aggressive; mitigated by caps and learning
2	Continuous measurement and AIMD-style concurrency control	Finds the sweet spot per server and per network	Oscillation; mitigated by thresholds and cooldowns
3	Work stealing by largest ETA, laggard split, connection recycling	Removes tail latency from slow edges	Complexity; mitigated by property-based tests
4	Per-host profiles	Skips slow start on repeat hosts	Stale data; mitigated by 30-day expiry
5	Zero-merge positional writes, bounded write queue	Disk never on the critical path; protects slow disks	HDD seek storms; mitigated by writer cap
6	Adaptive request windows (1-16 MiB)	Cheap ownership changes, reusable connections	Too many requests on high-latency links; window grows with speed

Note on HTTP/2: HTTP/2 multiplexes streams over a single TCP connection, so it does not bypass per-connection throttling. AJDM therefore defaults to HTTP/1.1 for bulk data and allows HTTP/2 only on hosts where it measured well.

11. Legal, ethics and compliance

- **Terms of service.** Many sites (including YouTube) restrict downloading content without permission. AJDM is a general tool; users are responsible for complying with the terms of the sites they use and with copyright law. First-run and About screens say so plainly.
- **No DRM circumvention.** Protected streams are detected and refused.
- **Third-party tools.** yt-dlp (Unlicense) and ffmpeg (use an LGPL build) run as separate processes. Ship license texts. Do not statically link GPL components.
- **Photograph and names.** The logo contains a family photograph. Keep the source image and permissions on record; do not license the logo under an open license that would allow others to reuse the photograph. Recommended: code under an open-source license, brand assets reserved.
- **Privacy.** No account, no telemetry by default, local-only data. Cookies are used only when the user opts in.
- **Trademarks.** AJDM does not use IDM's name or logo in the product. Comparative benchmarks name IDM factually and publish methodology.
- **IDM evaluation.** Benchmarks use a legally obtained IDM licence or trial per its terms. No reverse engineering of IDM binaries; behaviour is studied through observable network traces only.

12. Benchmark and validation plan

The goal to outshine IDM is only meaningful if it is measured the same way every time. The harness (ajdm-bench) automates the matrix below.

12.1 Method

- Same machine, same network, same file; alternate AJDM and IDM runs; cold disk cache; at least 10 runs each.
- Report median, p10 and p90 of wall-clock time and mean link utilisation; verify SHA-256 for every run.
- IDM runs on Windows (physical or VM). AJDM is also run on Linux and macOS for its own regression tracking.
- Publish raw CSVs and the scripts so anyone can reproduce.

12.2 Scenario matrix

#	Scenario	How produced	What it proves
B1	Unthrottled LAN, 2 GB	Local server	Engine overhead is negligible; link saturation
B2	Per-connection cap 2 MB/s	nginx limit_rate	Concurrency controller finds N x C
B3	Global cap 10 Mbit/s	tc / server cap	No wasted overhead when link-bound
B4	High latency (150 ms RTT)	tc netem delay	Window sizing and slow start
B5	1-3% packet loss	tc netem loss	Parallelism and recycling help under loss
B6	Flaky server (random disconnects)	FaultyRangeServer	Retry and resume robustness
B7	Real CDN large file	Public test files	Real-world behaviour
B8	YouTube 1080p and 2160p	Media pipeline	End-to-end including mux
B9	Mobile 4G hotspot	Physical phone hotspot	Target environment of primary users

12.3 Test gates for release

- 0 corrupted outputs in 1000 chaos runs; 200 of 200 kill-9 resumes bit-exact.
- Steal-operation invariants hold across 10,000 random interleavings.
- 24-hour soak shows no thread, memory or file-handle growth.
- Cross-OS CI green; installers install, launch, update and uninstall cleanly.

13. Risks and mitigations

Risk	Likelihood / Impact	Mitigation
YouTube/extractor changes break format listing	High / High	yt-dlp auto-update with checksum, fast hotfix path, clear error AJDM-4001
Servers throttle or ban many connections	Medium / Medium	Per-host cap, multiplicative decrease, Retry-After, per-host learning
JDK HttpClient limits prevent reaching target speed	Medium / High	HttpTransport interface; Netty implementation as planned escape hatch (decision gate in M2)
Not actually faster than IDM in some scenarios	Medium / Medium	Benchmark early (M2), iterate on thresholds, publish honest results
Legal/ToS complaints about video downloading	Medium / Medium	Clear user responsibility notice, no DRM bypass, neutral marketing
Solo capacity (study, job search, life)	High / High	Tight milestone slices, cut v1.1 features first, automate tests and releases
Antivirus false positives on installers	Medium / Medium	Code signing, reputation building, publish checksums
macOS notarization and Windows signing costs	Medium / Low	Budget early; ship Linux first if needed
Extension store review delays	Medium / Low	Submit early; offer sideload build
Disk and AV slowdowns on Windows	Medium / Medium	Bounded write queue backpressure; document exclusions
Data corruption bug	Low / Critical	Content-Range verification, If-Range, flush-before-commit, chaos suite as release gate

14. Milestones and timeline

Assumption: one developer working part-time (about 15 to 20 hours per week). Weeks are cumulative from project start. Dates are intentionally relative so the plan survives schedule changes.

Milestone	Weeks	Deliverables	Exit criteria
M0 Foundations	1-2	Repo, Gradle modules, CI on 3 OSes, FaultyRangeServer, probe, single-stream download, SQLite skeleton	Download a file end-to-end with correct hash in CI
M1 Segmented engine	3-6	Segments, workers, positional writer, checkpoints, resume, retry policy, CLI	Chaos suite passes at 100 runs; kill-9 resume works
M2 Adaptive speed	7-9	Concurrency controller, work stealing, laggard recycle, per-host profiles, bench harness, first IDM comparison	Benchmarks B1-B7 recorded; decision on HttpTransport
M3 Media	10-12	yt-dlp adapter, format model, presets, playlists, URL refresh, ffmpeg mux	4K video downloads and muxes bit-exact; playlist of 50 works
M4 Desktop UI	13-15	JavaFX main window, add dialog, media picker, details pane, settings, tray, themes	All Must UI requirements demoable
M5 Browser	16-17	Extension, native messaging shim, host registration, scheduler and post-actions	Chrome/Edge/Firefox intercept works fail-open
M6 Release	18-20	Installers, signing, About/dedication, docs, public benchmark report, beta feedback	Release gates in 12.3 satisfied; v1.0 tagged

15. Open questions and decisions needed

#	Question	Owner	Needed by
1	Project license for code (Apache-2.0 vs GPL-3.0) and separate brand-asset terms for the logo and photograph	Ethan	Before first public repo
2	Do we ship ffmpeg bundled, or download on first run? (size vs legal simplicity)	Ethan	M3
3	Windows and macOS code-signing budget and timing	Ethan	M5
4	Which exact IDM version and licence will be used for benchmarks	Ethan	M2
5	Default container for VP9/AV1 + Opus: mkv or webm	Ethan	M3
6	Website and domain for the project (for downloads, docs, benchmark report)	Ethan	M5

Appendix A. Settings catalogue (defaults)

Setting	Default	Notes
initialConnections	4	Range 1-8
maxConnectionsPerDownload	16	Hard cap 32
perHostCap / globalCap	16 / 64	Across all downloads
maxConcurrentDownloads	3	Per queue and global
minSegmentBytes	512 KiB	64 KiB alignment
windowBytes	Adaptive 1-16 MiB	<= 10 MiB for chunk-limited hosts
checkpointInterval	5 s	After fsync
connectTimeout / stallTimeout	10 s / 15 s	
maxRetriesPerWindow	10	Backoff 500 ms to 30 s, full jitter
controllerInterval	2 s	Gain thresholds +15% (slow start), +5% / -10% (hill climb)
maxHeight (media)	2160	Up to 4320
dataSaver	Off	Caps connections at 4
conflictPolicy	Rename	Rename / Overwrite / Ask
telemetry	Off	Opt-in crash reports only

Appendix B. Error codes

Code	Meaning	User action
AJDM-1001/1002	Unsupported scheme / invalid URL	Check the link
AJDM-2001	Not found (404)	Verify the link
AJDM-2002/2010	Access denied / credentials needed	Sign in or provide credentials
AJDM-2003/2004	No disk space / write error	Free space or choose another folder
AJDM-3001	File changed on server	Restart or keep partial
AJDM-3002	Retries exhausted	Check connection, retry
AJDM-3003	Resume unsupported	Download will restart if interrupted
AJDM-3004	Inconsistent server data	Retry; report host
AJDM-3005	Checksum mismatch	Re-download
AJDM-4001	Extraction failed	Update tools, retry
AJDM-4002	Link expired and refresh failed	Retry from page link
AJDM-4003	Mux failed	Open log; streams are kept
AJDM-4004	DRM-protected	Not supported
AJDM-5001	AJDM not running	Start AJDM

Appendix C. Glossary

Term	Definition
Segment	A byte range of a file owned by one worker connection.
Window	One HTTP Range request issued inside a segment.
Work stealing	Reassigning part of a slow segment's remaining bytes to a faster connection.
Laggard	A connection running far slower than the median of its peers.
Link-bound	Throughput limited by the user's network, not by the server.
Checkpoint	Fsync followed by a database commit of completed byte ranges.
Mux	Combining separate video and audio streams into one file without re-encoding.
DRM	Digital rights management; AJDM does not circumvent it.
IDM	Internet Download Manager, the Windows commercial download manager used as the benchmark.

AJDM — My brother. My legacy.